

University of Moratuwa

Faculty of Information Technology

CM3321 - Logic Programming and Artificial Cognitive Systems

Gemstone Identification Assistant

**A Rule-Based Expert System for Identifying Sri Lankan Gemstones and
Detecting Synthetics and Imitations**

Project Report

Name: Sansiluni J.A.S

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Implemented in SWI-Prolog 10.1.16

Abstract

Gem buyers, students and small traders in Sri Lanka frequently struggle to tell a valuable natural gemstone from a cheaper look-alike, a laboratory-grown synthetic or a glass imitation, while professional laboratory testing costs time and money. This project designs and implements the **Gemstone Identification Assistant**, a rule-based expert system written in SWI-Prolog that gives a quick, explained first opinion. The knowledge base contains 297 facts and 62 production rules, acquired from standard gemmological references and organised in four chained layers: species, variety, origin and treatment, and buying recommendation. The inference engine provides three reasoning modes over the same rule base: forward chaining (data-driven identification), backward chaining (goal-driven verification of a claim, implemented as a meta-interpreter that builds proof trees) and an interactive consultation that asks the user only for the facts it needs. MYCIN-style certainty factors express the confidence of each conclusion and turn incomplete data into a ranked list of candidates. A full explanation facility answers *How*, *Why* and *Why not*, both as a technical trace and in plain language, and animated reasoning diagrams show how each chaining method worked for the user's own input. A web interface, a console interface and a natural-language query box built with a Definite Clause Grammar make the system accessible to non-experts. Eighteen documented test cases and 41 automated unit tests pass. The report closes with a discussion of how the system models expert cognition, a comparison with machine-learning approaches, and an honest account of its limitations.

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1. Introduction and Problem Definition

1.1 Background

Sri Lanka is one of the world's most important sources of coloured gemstones. Sapphire, ruby, spinel, chrysoberyl (including alexandrite and cat's-eye), garnet, zircon, topaz, tourmaline, quartz and moonstone are mined and traded in large numbers, particularly around Ratnapura. The same trade attracts imitations (coloured glass, cubic zirconia), synthetics (laboratory-grown corundum and spinel) and undisclosed treatments such as heating. Gemmologists separate these materials with a small set of physical and optical tests: refractive index, specific gravity, optic character, optical phenomena and the inclusions seen under magnification [1]-[4]. These tests follow well-documented reasoning, which makes gem identification a good candidate for a rule-based expert system.

1.2 Problem statement

Buyers, students and small traders often cannot tell a valuable natural gem from a cheaper look-alike or a fake, and deception is a known problem in the local trade. Accredited laboratories such as the National Gem and Jewellery Authority (NGJA) of Sri Lanka [7] give reliable answers but cost money and time, so they are not practical for every stone a buyer is offered. There is a need for a tool that takes the observations an informed buyer or student can realistically make and returns a quick, **explained** first opinion: what the stone probably is, whether it is likely natural, synthetic or an imitation, how confident that opinion is, and what the buyer should do next.

1.3 Aim and objectives

The aim is to build an expert system that identifies common Sri Lankan gemstones and detects synthetics and imitations, and that explains its reasoning. The objectives are:

1. O1 - Acquire gemmological knowledge from recognised sources and document how it became facts and rules.
2. O2 - Represent the knowledge as Prolog facts and production rules, with at least 20 of each.
3. O3 - Implement both forward and backward chaining over the same rule base.
4. O4 - Provide an explanation facility that shows user input, facts, rules fired, reasoning and conclusion.
5. O5 - Handle incomplete data and express the confidence of conclusions.
6. O6 - Provide a user-friendly interface for non-experts.
7. O7 - Test the system systematically and document it with a user manual.

1.4 What the system specialises in

The system specialises in ten gem species that are common in Sri Lanka (corundum, spinel, garnet, chrysoberyl, zircon, topaz, tourmaline, quartz, orthoclase feldspar and beryl), eighteen named varieties (for example ruby, blue sapphire, padparadscha, alexandrite, cat's-eye chrysoberyl, moonstone, hessonite and aquamarine) and the two most common imitation materials, glass and cubic zirconia. It reasons from six observable properties: refractive index (RI), specific gravity (SG), optic character, body colour, optical phenomenon and the dominant inclusion type.

1.5 Conclusions the system provides

Table 1. Conclusions produced by the system

Conclusion	Description and example
Species	The mineral species, e.g. <i>corundum</i> , from optic character, RI and SG.
Variety	The trade variety within the species, e.g. corundum + blue gives <i>blue sapphire</i> ; chrysoberyl + colour change gives <i>alexandrite</i> .
Imitation	Glass or cubic zirconia, e.g. an isotropic stone with a glass RI and gas bubbles.
Origin and treatment	Likely natural, synthetic (laboratory-grown), imitation, glass-or-synthetic, laboratory test needed, or not determined; and probable heat treatment.
Confidence	A certainty factor for every derived fact (e.g. <i>blue sapphire 90%</i>), a confidence level (strong match / possible match / no match), and a ranked candidate list when data is missing (e.g. <i>Spinel 52%, Garnet 52%</i>).
Recommendation	Practical advice, e.g. <i>"Features are consistent with a natural stone. Obtain an NGJA certificate before purchase."</i>
Explanation	How the conclusion was reached, why a question is being asked, and why an alternative was rejected.

1.6 Scope and limitations

The system is a decision-support tool for a first opinion; it is **not** a replacement for laboratory certification. The knowledge base covers ten species and two imitations. Heat treatment can only be suggested from a visible indicator (discoïd fractures), not proven. The system assumes the user can take basic readings (a refractometer, a hydrostatic balance and a polariscope) but works with partial data. Section 7.3 discusses the limitations in detail.

2. Knowledge Engineering

Knowledge engineering was carried out as an iterative process (Figure 1). The domain was studied from recognised sources, individual knowledge statements were extracted into a knowledge acquisition table, encoded as Prolog facts and rules, tested, and refined when tests revealed gaps.

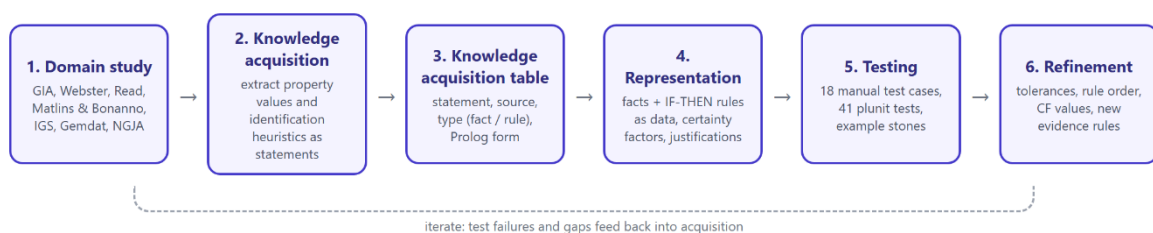


Figure 1. Knowledge engineering process used in the project

2.1 Domain study: gemmology concepts

Refractive index (RI) measures how strongly a gem slows and bends light. It is read on a gem refractometer and is the single most diagnostic test for most transparent gems [3]. Corundum reads about 1.762-1.770, quartz about 1.544-1.553 [1], [3]. Standard refractometers cannot read above about 1.81, so very high-RI materials such as high zircon and cubic zirconia show as "over the limit".

Specific gravity (SG) is the density of the stone relative to water, measured by hydrostatic weighing. It separates gems with similar RI; corundum (about 4.00) is much heavier than spinel (about 3.60) [2].

Optic character describes how light passes through the crystal. Isotropic (singly refractive) materials such as spinel, garnet, glass and cubic zirconia transmit light as a single ray. Anisotropic (doubly

refractive) materials split it in two; they are *uniaxial* (one optic axis: corundum, quartz, beryl, tourmaline, zircon) or *biaxial* (two axes: chrysoberyl, topaz, feldspar) [2], [3]. It is determined with a polariscope and conoscope. **Birefringence** is the difference between the two RI readings of a doubly refractive gem.

Optical phenomena are special light effects: asterism (a star) and chatoyancy (a cat's-eye) caused by aligned needle-like inclusions, colour change (typical of alexandrite) and adularescence (the floating sheen of moonstone) [1].

Inclusions are the gemmologist's main field evidence for origin. Rutile "silk" and mineral crystals indicate natural growth; curved growth lines (striae) and round gas bubbles indicate flame-fusion (Verneuil) synthetics; swirl marks and bubbles indicate glass; discoid fractures around crystals indicate heat treatment [1], [4], [5].

Synthetic versus imitation. A synthetic has the same chemistry and properties as the natural gem but was grown in a laboratory; an imitation merely looks similar and has different properties [4]. The distinction matters because a synthetic ruby is still corundum and passes the species tests, so only inclusion evidence can reveal it.

Table 2 summarises the property facts acquired for each material. These are refractometer and hydrostatic *reading ranges*, which are slightly wider than textbook mean values so that normal variation within a species is covered.

Table 2. Property facts held in the knowledge base (generated from knowledge_base.pl)

Material	RI range	SG range	Optic character	Birefringence	Hardness
Corundum	1.757 - 1.780	3.95 - 4.10	uniaxial	0.008	9
Spinel	1.712 - 1.736	3.55 - 3.70	isotropic	-	8
Garnet	1.730 - 1.890	3.57 - 4.30	isotropic	-	7
Chrysoberyl	1.740 - 1.760	3.70 - 3.78	biaxial	0.009	8.5
Zircon	1.810 - 1.990	4.60 - 4.80	uniaxial	0.059	7.5
Topaz	1.606 - 1.644	3.49 - 3.57	biaxial	0.009	8
Tourmaline	1.614 - 1.666	3.00 - 3.26	uniaxial	0.018	7.5
Quartz	1.540 - 1.558	2.60 - 2.70	uniaxial	0.009	7
Feldspar	1.515 - 1.530	2.55 - 2.63	biaxial	0.006	6
Beryl	1.564 - 1.602	2.63 - 2.92	uniaxial	0.006	7.5
Glass	1.440 - 1.770	2.20 - 4.50	isotropic	-	5.5
Cubic zirconia	2.150 - 2.180	5.60 - 6.00	isotropic	-	8.5

2.2 Domain study: expert system concepts and related work

An expert system captures the specialist knowledge of human experts so that non-experts can apply it [10], [11]. The classic architecture separates a **knowledge base** (domain facts and heuristics), an **inference engine** (a domain-independent reasoning procedure), **working memory** (the facts of the current case) and a **user interface** with an **explanation facility** [11]. This separation means the knowledge can be extended without changing the reasoning code.

Production rules (IF conditions THEN conclusion) are the most widely used knowledge representation in expert systems because they match the way experts state heuristics and because each rule can be explained individually [10]. Two inference strategies operate on production rules [11], [17]:

- **Forward chaining** is data-driven. Starting from the known facts, the engine repeatedly finds a rule whose conditions are all satisfied, adds its conclusion to working memory, and stops when no rule can add anything new. When several rules are applicable, a **conflict resolution** strategy (for example rule order or specificity) chooses which fires, and **refractoriness** prevents the same rule instance firing twice.
- **Backward chaining** is goal-driven. Starting from a hypothesis, the engine looks for rules that conclude it and tries to prove their conditions as sub-goals, recursively, until it reaches known facts. Prolog's own execution model is backward chaining with depth-first search and backtracking [12].

The best-known rule-based expert system, **MYCIN**, diagnosed bacterial infections using about 450 backward-chained rules [8]. It introduced two ideas this project adopts. The first is **certainty factors**: each rule carries a measure of belief, and evidence from several rules is combined incrementally [9]. The second is a consultation dialogue that asks the user only for the information a rule needs and can answer "why?" by showing the rule being tried [8], [9]. Later work such as XPLAIN argued that good explanations must justify the reasoning, not merely replay it [20]. That motivated the plain-language explanations and the source-linked rule justifications in this system.

Bratko [12] and Merritt [13] show how to build expert system shells in Prolog: rules are stored as data, a meta-interpreter proves goals and records a proof tree for "how" explanations, and questions are asked lazily during proof. Sterling and Shapiro [14] describe the meta-interpreter technique in depth, and Pereira and Warren [15] introduced Definite Clause Grammars (DCGs), which this project uses for its natural-language interface. SWI-Prolog [16] provides the HTTP and HTML libraries used for the web interface.

[Add here 2-3 published papers on gemstone identification systems or rule-based expert systems in mineralogy/gemmology found on Google Scholar, and state how this system differs, e.g. its Sri Lankan focus, certainty factors, three reasoning modes and plain-language explanation facility. Add them to the reference list.]

2.3 Knowledge acquisition

Knowledge was acquired from documentary sources: the GIA Gem Encyclopedia [1], Webster's *Gems* [2], Read's *Gemmology* [3], Matlins and Bonanno's *Gem Identification Made Easy* [4], the International Gem Society [5], the Gemdat database [6] and the NGJA [7]. Each usable piece of knowledge was written as a short statement, classified as a fact or a rule, and encoded in Prolog. Every rule in the knowledge base stores its justification and source in `rule_info(Id, Justification, Source)`, so the acquisition trail is kept inside the system itself and appears in every explanation. Table 3 shows a representative selection of the 41 main knowledge statements; the complete rule base with justifications is in Appendix A.

Three knowledge engineering decisions were made during acquisition:

1. **Reading ranges plus tolerance.** Published ranges were stored as reading ranges, and the engine adds a measurement tolerance of ± 0.005 for RI and ± 0.03 for SG to allow for instrument and operator error. A reading that fits only because of the tolerance is accepted with reduced certainty (Section 2.5).
2. **Layered rules.** Knowledge was organised by the question it answers (species, variety, origin, advice), so that conclusions of one layer become conditions of the next.

3. **Explicit uncertainty.** Where sources describe evidence as typical rather than conclusive (for example, fingerprints usually indicate a natural stone, but flux-grown synthetics show similar features), the rule was given a lower certainty factor rather than being omitted.

Table 3. Knowledge acquisition table (representative entries; KE = knowledge engineer decision)

#	Knowledge statement (from source)	Source	Type	Prolog form
K1	Corundum reads about 1.762-1.770 on the refractometer.	[1]	Fact	ri_range(corundum, 1.757, 1.780).
K2	Corundum has SG about 4.00.	[1]	Fact	sg_range(corundum, 3.95, 4.10).
K3	Corundum is doubly refractive and uniaxial.	[1]	Fact	optic(corundum, uniaxial).
K4	Spinel is singly refractive, RI about 1.718, SG about 3.60.	[2]	Fact	optic(spinel, isotropic). ri_range(spinel, 1.712, 1.736).
K5	Garnets are singly refractive, RI 1.73-1.89, SG 3.6-4.3 depending on type.	[2]	Fact	ri_range(garnet, 1.730, 1.890).
K6	Chrysoberyl is biaxial, RI about 1.746-1.755, SG about 3.73.	[1]	Fact	optic(chrysoberyl, biaxial).
K7	High zircon is uniaxial with very high RI and SG about 4.7.	[6]	Fact	sg_range(zircon, 4.60, 4.80).
K8	Topaz is biaxial, RI about 1.61-1.64, SG about 3.53.	[1]	Fact	ri_range(topaz, 1.606, 1.644).
K9	Quartz is uniaxial, RI 1.544-1.553, SG 2.65.	[3]	Fact	sg_range(quartz, 2.60, 2.70).
K10	Moonstone (orthoclase) is biaxial, RI about 1.52, SG about 2.57.	[3]	Fact	ri_range(feldspar, 1.515, 1.530).
K11	Beryl is uniaxial, RI about 1.57-1.60, SG about 2.7.	[6]	Fact	optic(beryl, uniaxial).
K12	Glass is singly refractive with a wide range of RI.	[5]	Fact	ri_range(glass, 1.440, 1.770).
K13	Cubic zirconia is isotropic with SG 5.6-6.0.	[1]	Fact	sg_range(cubic_zirconia, 5.60, 6.00).
K14	Corundum occurs in red, pink, orange, yellow, green, blue, purple and colourless.	[1]	Fact	colour(corundum, blue).
K15	Chrysoberyl can show chatoyancy and colour change.	[1]	Fact	phenomenon(chrysoberyl, colour_change).
K16	Readings vary slightly with instrument and operator.	KE	Fact	tolerance(ri, 0.005). tolerance(sg, 0.03).
K17	A uniaxial stone with corundum RI and SG is corundum.	[1]	Rule	rule(r1, [optic(uniaxial), ri_in(corundum), sg_in(corundum)], species(corundum)).
K18	A singly refractive stone with RI about 1.72 and SG about 3.60 is spinel.	[2]	Rule	rule(r2, [optic(isotropic), ri_in(spinel), sg_in(spinel)], species(spinel)).
K19	An isotropic stone with a glass RI and gas bubbles that matches no species is glass.	[5]	Rule	rule(r11, [optic(isotropic), ri_in(glass), inclusion(gas_bubbles), not(species(_))], imitation(glass)).

K20	Swirl marks are typical of moulded glass.	[4]	Rule	rule(r12, [inclusion(swirl_marks), not(species(_))], imitation(glass)).
K21	An isotropic stone with SG 5.6-6.0 is cubic zirconia.	[1]	Rule	rule(r13, [optic(isotropic), sg_in(cubic_zirconia)], imitation(cubic_zirconia)).
K22	When data is missing, every gem consistent with all supplied properties remains possible.	KE	Rule	rule(r14, [partial_data, ..., optic_ok(G), ri_ok(G), sg_ok(G), colour_ok(G), phenomenon_ok(G)], candidate(G)).
K23	A matching RI is the strongest single test.	[3]	Rule	rule(r16, [candidate(G), not(missing(ri)), ri_ok(G)], candidate(G)).
K24	Red corundum is ruby.	[1]	Rule	rule(r21, [species(corundum), observed_colour(red), not(phenomenon(star))], variety(ruby)).
K25	Blue corundum is blue sapphire.	[1]	Rule	rule(r24, [species(corundum), observed_colour(blue), not(phenomenon(star))], variety(blue_sapphire)).
K26	Non-red corundum with asterism is star sapphire.	[1]	Rule	rule(r23, [species(corundum), phenomenon(star), not(observed_colour(red))], variety(star_sapphire)).
K27	Pinkish-orange Sri Lankan corundum is traded as padparadscha.	[1]	Rule	rule(r26, [species(corundum), observed_colour(orange), ...], variety(padparadscha)).
K28	Chrysoberyl with a daylight/incandescent colour change is alexandrite.	[1]	Rule	rule(r29, [species(chrysoberyl), phenomenon(colour_change)], variety(alexandrite)).
K29	Orthoclase feldspar showing adularescence is moonstone.	[1]	Rule	rule(r31, [species(feldspar), phenomenon(adularescence)], variety(moonstone)).
K30	Orange grossular garnet is hessonite.	[1]	Rule	rule(r34, [species(garnet), observed_colour(orange)], variety(hessonite)).
K31	Curved striae in corundum prove flame-fusion synthesis; natural corundum grows in straight lines.	[1]	Rule	rule(r41, [species(corundum), inclusion(curved_lines)], origin(synthetic)).
K32	Round gas bubbles in corundum indicate a flame-fusion synthetic.	[1]	Rule	rule(r42, [species(corundum), inclusion(gas_bubbles)], origin(synthetic)).
K33	Gas bubbles in an unidentified stone indicate glass or a synthetic.	[5]	Rule	rule(r44, [inclusion(gas_bubbles), not(species(_)), not(imitation(_))], origin(glass_or_synthetic)).
K34	Rutile silk is a natural inclusion in corundum.	[1]	Rule	rule(r46, [species(corundum), inclusion(silk)], origin(likely_natural)).

K35	Included mineral crystals indicate natural growth.	[4]	Rule	<code>rule(r47, [species(_), inclusion(crystals)], origin(likely_natural)).</code>
K36	Fingerprints are usually natural, but flux synthetics show similar features.	[4]	Rule (CF 0.6)	<code>rule(r48, [species(_), inclusion(fingerprints)], origin(likely_natural)).</code>
K37	Discoïd fractures form when corundum is heated.	[1]	Rule	<code>rule(r49, [species(corundum), inclusion(discoïd_fractures)], treatment(likely_heated)).</code>
K38	Natural corundum is rarely clean; a clean stone needs a laboratory test.	[5]	Rule	<code>rule(r51, [species(corundum), inclusion(none)], origin(needs_lab_test)).</code>
K39	Natural stones should be certified before purchase (NGJA).	[7]	Rule	<code>rule(r61, [origin(likely_natural)], advice(get_certificate)).</code>
K40	Treatments must be disclosed to the buyer.	[7]	Rule	<code>rule(r67, [treatment(likely_heated)], advice(disclose_heat)).</code>
K41	Alexandrite must be confirmed under daylight and incandescent light.	[1]	Rule	<code>rule(r68, [variety(alexandrite)], advice(check_colour_change)).</code>

[Check every value and statement in Tables 2 and 3 against the cited source before submission, and adjust the source column if needed.]

2.4 Knowledge representation

The knowledge is represented as **Prolog facts plus production rules**. Prolog suits this because it is declarative (facts and rules are written as logical statements), has pattern matching through unification, and has built-in backtracking, which is already a form of backward chaining [12].

The key design decision is that every rule is stored as a **data term**, `rule(Id, Conditions, Conclusion)`, rather than as a Prolog clause. This has three benefits. The same rule base can be interpreted by both a forward chainer and a backward chainer. Every rule has an identifier that can be shown in explanations. Rules can be inspected, listed on the knowledge base page and exported for documentation. Conditions are either facts in working memory (for example `observed_colour(red)`), conclusions of other rules (for example `species(corundum)`), computed tests (for example `ri_in(corundum)`, which compares the RI reading with the stored range), or negations `not(c)` evaluated by negation as failure.

Numeric ranges are kept in facts rather than inside rules: rule r1 states `ri_in(corundum)` and the engine looks up `ri_range(corundum, Min, Max)`. This keeps the property data in one place and lets the explanation say "RI 1.765 lies within the corundum range 1.757-1.780". Table 4 documents the predicates.

Table 4. Main predicates of the knowledge base and working memory

Predicate	Meaning
<code>gem(Name)</code>	A gem species known to the system (10).
<code>imitation_material(Name)</code>	An imitation material (glass, cubic zirconia).
<code>ri_range(Gem, Min, Max)</code>	Refractive index reading range.
<code>sg_range(Gem, Min, Max)</code>	Specific gravity range.
<code>optic(Gem, Type)</code>	Optic character: isotropic, uniaxial or biaxial.

hardness(Gem, H), birefringence(Gem, B)	Mohs hardness and birefringence (information only).
colour(Gem, C), typical_colour(Gem, C)	Possible colours; colours most common in the trade.
phenomenon(Gem, P)	Optical phenomena the species can show.
inclusion_type(I, Description)	Inclusion types the user can report.
rule(Id, Conditions, Conclusion)	Production rule (62).
rule_cf(Id, CF)	Certainty factor of a rule.
rule_info(Id, Text, Source), source(Id, Citation)	Justification and literature source of each rule.
advice_text(Key, Text)	Wording of each recommendation.
fact(F)	Working memory: observed and derived facts of the current case (dynamic, thread-local).
fired(Id, Conds, Concl)	Log of fired rules, used for the How explanation.
cf(F, CF), fired_cf/6	Certainty of each derived fact and of each rule firing.

The rules are grouped into four layers (Figure 2). Layer 1 identifies the species from the three core measurements, detects imitations, and, when measurements are missing, produces ranked candidates. Layer 2 names the variety. Layer 3 judges origin and treatment from inclusions. Layer 4 turns conclusions into advice. Because each layer uses the conclusions of the one above as conditions, a single identification typically chains three or four rules.

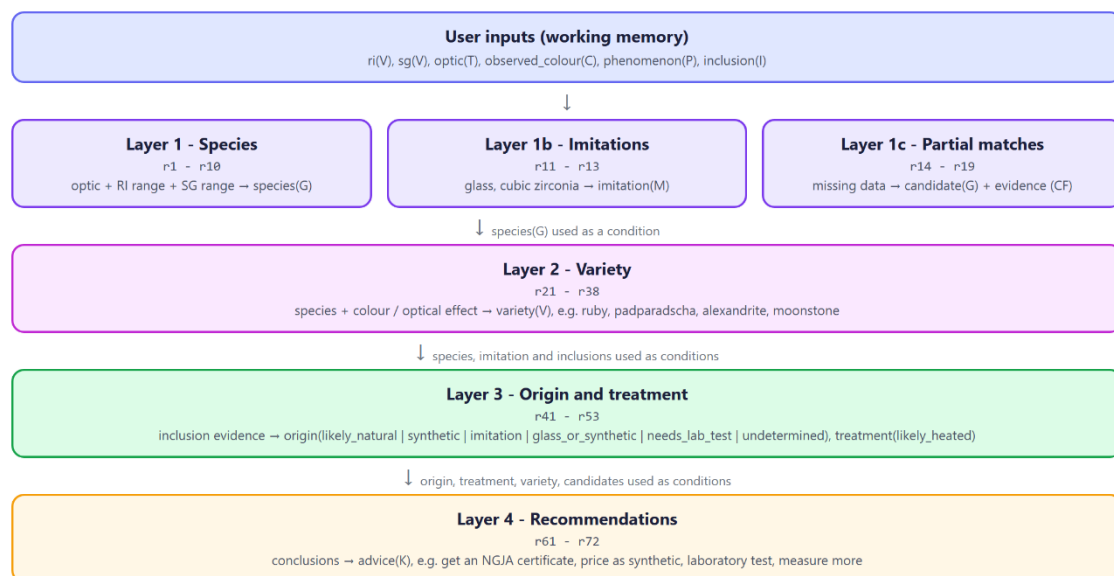


Figure 2. Rule layers and how conclusions chain between them

The following excerpt shows representative rules from each layer:

```
% Layer 1: species
rule(r1, [optic(uniaxial), ri_in(corundum), sg_in(corundum)], species(corundum)).
```

```

% Layer 2: variety
rule(r24, [species(corundum), observed_colour(blue), not(phenomenon(star))],
    variety(blue_sapphire)).

% Layer 3: origin
rule(r46, [species(corundum), inclusion(silk)], origin(likely_natural)).

% Layer 4: recommendation
rule(r61, [origin(likely_natural)], advice(get_certificate)).

```

Alternative representations were considered. A **decision tree** would force a fixed order of questions and could not use whatever subset of data the user has. **Frames** would describe each gem well but do not naturally express the heuristic IF-THEN knowledge about inclusions and origin. Production rules express both kinds of knowledge, support both chaining directions and give natural explanations, so they were chosen.

2.5 Representing uncertainty: certainty factors

Gem identification is rarely certain: ranges of different species overlap slightly, colour names are judged by eye, and some inclusions occur in both natural and synthetic stones. Following MYCIN [9], each rule has a certainty factor between 0 (no belief) and 1 (certain), stored as `rule_cf(Id, CF)`. The certainty of a conclusion is the rule's certainty multiplied by that of its weakest condition:

$$CF(\text{conclusion}) = CF(\text{rule}) \times \min(CF(\text{condition}_1), \dots, CF(\text{condition}_n))$$

User inputs have CF 1.0. A reading inside a gem's published range has CF 1.0, while a reading that fits only because of the measurement tolerance has CF 0.7. When a second rule supports a fact that is already believed, the two certainties are combined:

$$CF_{\text{combined}} = CF_1 + CF_2 \times (1 - CF_1)$$

Table 5 summarises the values chosen for each rule group; individual values are listed in Appendix A.

Table 5. Certainty factor ranges by rule group

Rule group	Rules	CF range
Species (layer 1)	r1 - r10	0.90 - 0.95
Imitations (layer 1b)	r11 - r13	0.80 - 0.95
Partial-match base and evidence (layer 1c)	r14 - r19	0.15 - 0.50
Varieties (layer 2)	r21 - r38	0.70 - 0.95
Origin and treatment (layer 3)	r41 - r53	0.60 - 1.00
Recommendations (layer 4)	r61 - r72	1.00 - 1.00

The values encode the knowledge engineer's judgement of how conclusive each piece of evidence is. Species rules are high (0.90-0.95) because three independent measurements agree, but not 1.0 because ranges can overlap. Colour-based varieties are lower where the colour boundary is subjective (padparadscha 0.70). Origin rules range from 0.95 for curved striae, which the literature treats as diagnostic of flame-fusion synthesis [1], down to 0.60 for fingerprints, which can also occur in flux-grown synthetics [4]. The partial-match evidence rules are deliberately weak (0.15-0.50) so that incomplete data never produces a falsely confident answer.

[Be ready to justify these values in the viva; adjust them if your own reading of the sources suggests different weights.]

3. System Design and Implementation

3.1 Architecture

The system follows the classic expert system architecture (Figure 3). The knowledge base (knowledge_base.pl) contains only domain knowledge and no reasoning code. The inference engine (engine.pl) contains only domain-independent reasoning and explanation code. Adding a new gem therefore means adding facts and rules; no engine code changes. Working memory is declared thread_local, so each web request, which runs in its own thread, has a private copy and several users cannot interfere with each other.

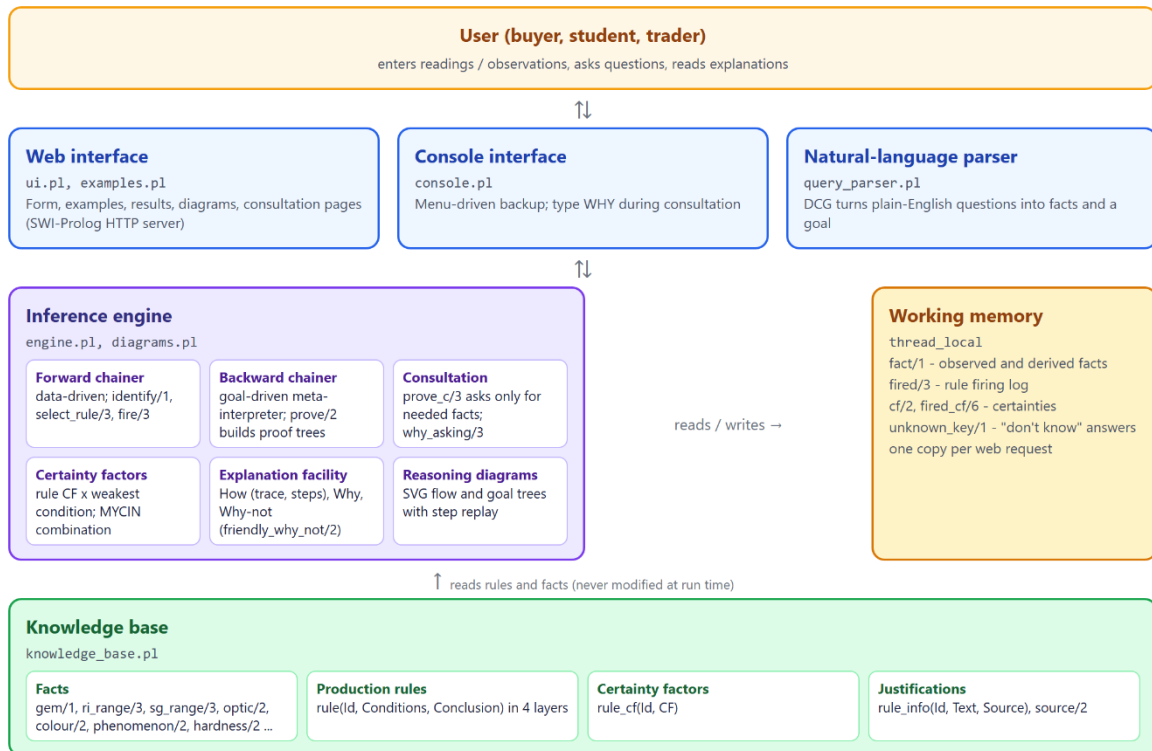


Figure 3. System architecture

Table 6. Source files

File	Responsibility
knowledge_base.pl	Facts, 62 production rules, certainty factors, justifications and sources.
engine.pl	Working memory, forward chaining, backward chaining meta-interpreter, consultation, certainty factors, How/Why/Why-not explanations.
diagrams.pl	Generates the forward-chaining flow diagram and the backward-chaining goal tree as SVG.
query_parser.pl	DCG that parses plain-English questions.
ui.pl, examples.pl	Web interface (SWI-Prolog HTTP server), example stones and gem pictures.
console.pl	Console interface.
tests.pl	41 automated tests (plunit).

main.pl	Loads all modules.
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3.2 Forward chaining (Identify mode)

In Identify mode the user's observations are asserted into working memory and the engine runs the recognise-act cycle until it reaches a fixed point:

```
forward_chain :-
    (   select_rule(Id, Conds, Concl)      % recognise
    ->  fire(Id, Conds, Concl),           % act
        forward_chain                    % next cycle
    ;   true                             % no rule can fire: stop
    ).

select_rule(Id, Conds, Concl) :-
    rule(Id, Conds, Concl),              % candidate rule, in knowledge-base order
    all_true(Conds),                     % every condition holds in working memory
    ground(Concl),
    \+ fired(Id, _, Concl).              % refractoriness
```

Conflict resolution is by rule order: in each cycle the first applicable rule in the knowledge base fires. Because the rules are ordered by layer, all species rules fire before variety rules, which fire before origin and advice rules. This ordering is what makes the negative conditions safe: for example, rule r11 (glass) requires `not(species(_))`, and by the time it is considered every species rule has already had the chance to fire. **Refractoriness** prevents a rule from firing twice for the same conclusion; a different rule may still support an existing fact, in which case its certainty is combined into that fact. `fire/3` asserts the conclusion, records the firing in `fired/3` for the explanation facility, and stores the certainty calculation in `fired_cf/6`.

For the input RI 1.765, SG 4.00, uniaxial, blue, silk, the engine fires r1 (species corundum), r24 (blue sapphire), r46 (likely natural) and r61 (certificate advice), then stops because no further rule can add a new fact.

3.3 Backward chaining (Verify mode)

In Verify mode the user asks whether the stone is a particular gem, for example "is it a ruby?". The goal `variety(ruby)` is proved by a **meta-interpreter**, a Prolog program that interprets the rule base and builds a proof tree as it goes [12], [14]:

```
prove(not(C), negated(C))    :- !, \+ prove(C, _).          % negation as failure
prove(C, test(C))           :- test_condition(C), !, test(C). % computed test
prove(C, given(C))          :- input_fact(C), !, fact(C).    % user input
prove(C, by(Id, C, Proofs)) :-                               % use a rule
    rule(Id, Conds, C),
    prove_all(Conds, Proofs).
```

To prove a goal the engine selects a rule whose conclusion unifies with it and recursively proves its conditions as sub-goals, backtracking to the next rule if a condition fails. The proof tree `by(r21, variety(ruby), [by(r1, species(corundum), [...]), given(observed_colour(red)), negated(phenomenon(star))])` is the basis of the How explanation and the backward-chaining diagram. When the goal fails, `why_not/3` examines every rule that could have concluded it, finds the **first failed**

condition of each, and, if that condition is itself a derived fact, explains recursively why that sub-goal failed.

3.4 Interactive consultation (Consult mode)

The third mode reproduces the MYCIN-style consultation [8]. `prove_c/3` is a variant of the meta-interpreter that keeps a **stack of the rules being tried**. When a condition needs a property the user has not yet supplied, it throws `ask(Property, Stack)`:

```
ensure_known(Stack, K) :-
    input_key(K, F),
    (   fact(F)          -> true           % already answered
    ;   unknown_key(K) -> true           % user said "don't know"
    ;   throw(ask(K, Stack))             % ask the user now
    ).
```

The interface asks the question and restarts the consultation with the new answer. Questions are therefore asked only when, and only if, a rule being tested needs them. When the goal is "find the species", backward chaining naturally performs **hypothesise-and-test**: it tries `r1` (is it corundum?), asks for the optic character, and if the answer rules corundum out it abandons that hypothesis and moves on to `r2` (is it spinel?) without asking anything about corundum again. The stack gives the answer to "Why are you asking this?" (Section 4.3).

3.5 Certainty factors and ranked candidates

When RI, SG or optic character is missing, no species rule can fire. Rule `r14` then makes every gem that is *consistent* with all supplied properties a candidate with a low base certainty (0.20), and evidence rules `r15`-`r19` add certainty for each supplied property that positively matches. For a red, singly refractive stone the calculation for spinel is:

<code>r14</code>	consistent with all inputs	CF = 0.20
<code>r15</code>	optic character matches (rule 0.30)	CF = 0.20 + 0.30 x (1 - 0.20) = 0.44
<code>r19</code>	red is a typical colour (rule 0.15)	CF = 0.44 + 0.15 x (1 - 0.44) = 0.524

Garnet receives the same evidence, so the system ranks "Spinel 52%, Garnet 52%". This tie is honest: these observations genuinely cannot separate the two, and the system advises measuring RI and SG. With uniaxial and blue it ranks corundum, zircon and beryl (52%) above tourmaline (44%), because blue is not a typical tourmaline colour. An evidence rule's own conclusion is excluded from its weakest-condition calculation, since a belief is not evidence for itself.

3.6 Natural-language interface

Users can type a question such as "Is my stone a sapphire if the RI is 1.765 and it is blue?". The question is tokenised and parsed by a Definite Clause Grammar [15] in `query_parser.pl`. The grammar recognises goal phrases ("is my stone a", "could it be"), property phrases ("RI is 1.765", "singly refractive", "curved growth lines") and gem names, and skips words it does not know, so word order is flexible:

```
items([I|Is])      --> item(I), !, items(Is).
items([skip(W)|Is]) --> [W], !, items(Is).      % unknown word: skip it
items([])          --> [].
```

```

item(goal(T)) --> goal_phrase(T).
item(fact(F)) --> fact_phrase(F).

goal_phrase(T) --> [is], stone_ref, article, target(T).
fact_phrase(ri(N)) --> ri_word, link, number(N).

```

The example question is parsed as the goal `verify(blue_sapphire)` with facts `[ri(1.765), observed_colour(blue)]` and answered by backward chaining. Gem names are matched longest first, so "star sapphire" is not mistaken for "sapphire".

3.7 User interface

The main interface is a web application served by SWI-Prolog's built-in HTTP library, so nothing beyond SWI-Prolog needs to be installed. The home page introduces the system, explains the two chaining methods in plain terms, and provides a form in which each field explains what the property means and how to measure it (Figures 4 and 5). Result pages present the conclusion, certainty and origin as badges, the recommendation, a reasoning diagram and the plain-language reasoning steps (Figure 6). A menu-driven console interface provides the same functions as a backup. Full instructions are in the separate User Manual.

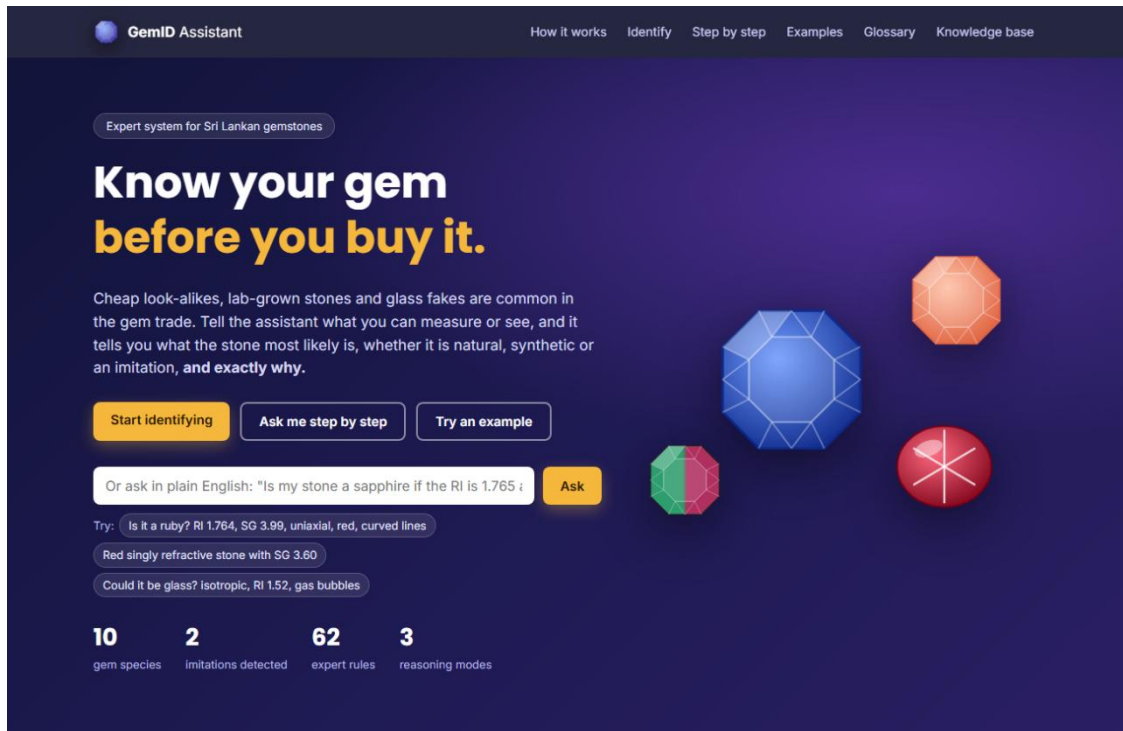


Figure 4. Home page introducing the system, with the plain-English question box

Describe your stone

Every field is optional. The more you fill in, the more confident the answer.

1 Instrument readings

Refractive index (RI)

How strongly the stone bends light. Each gem has its own range.

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)

How heavy the stone is compared with the same volume of water.

HOW Weigh the stone in air and in water (hydrostatic balance).

Optic character

Whether light passes through as one ray or is split in two.

HOW Turn the stone between crossed filters in a polariscope.

2 What you can see

Colour

The main body colour of the stone.

HOW Look at it in daylight against a white background.

Optical effect

Special light effects such as a star or a cat's-eye.

HOW Look under a single light source and rotate the stone.

Inclusions

Tiny features inside the stone that reveal how it formed.

HOW Examine it with a 10x jeweller's loupe.

Identify my stone

Find the best match using every rule in the knowledge base.

Check a claim

Test whether the stone could be one particular gem.

Ask me step by step

The system asks only the questions it needs, and you can ask it why.

[Clear the form](#)

Example stones to try

Figure 5. Input form with an explanation of each property and the three reasoning modes

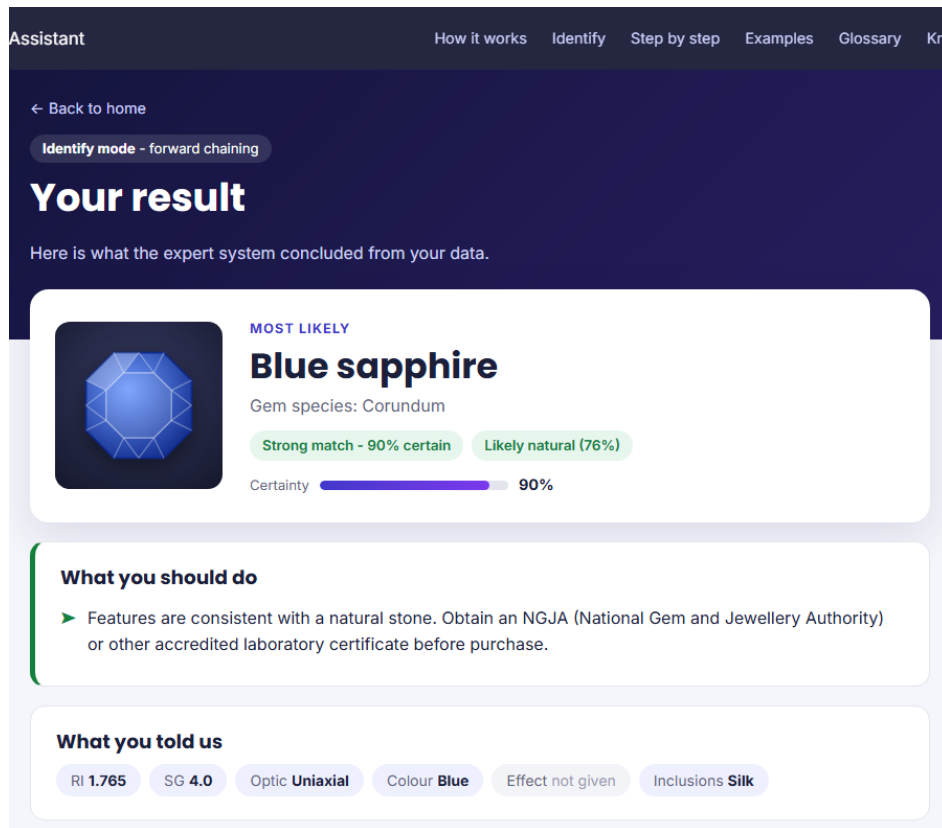


Figure 6. Identification result for a natural blue sapphire

3.8 Logic programming concepts used

Table 7. Where the system uses core Prolog concepts

Concept	Where and how it is used
Unification	Matching rule conclusions to goals in <code>prove/2</code> (engine.pl, line 347); a rule with variables, e.g. <code>r14 candidate(G)</code> , is instantiated once per matching gem.
Backtracking	<code>select_rule/3</code> (line 164) backtracks through rules and variable bindings until all conditions hold; <code>prove/2</code> backtracks to the next rule when a sub-goal fails.
Recursion	<code>forward_chain/0</code> (line 155) recurses once per cycle; <code>prove_all/2</code> , <code>why_not/3</code> (line 359) and the DCG <code>items//1</code> recurse over lists and trees.
Negation as failure	<code>test(not(C)) :- \+ holds(C)</code> (line 107) and <code>prove(not(C), ...)</code> (line 344) implement conditions such as <code>not(phenomenon(star))</code> .
Dynamic database	Working memory <code>fact/1</code> , <code>fired/3</code> , <code>cf/2</code> is changed with <code>assertz/retract</code> in <code>load_inputs/1</code> (line 91) and <code>fire/3</code> (line 171); declared <code>thread_local</code> (lines 37-41).
Meta-interpreter	<code>prove/2</code> and <code>prove_c/3</code> interpret rules stored as data and build proof trees.
Exceptions	The consultation throws <code>ask(K, Stack)</code> from inside the proof (line 265) and catches it in <code>consult/4</code> (line 236).
Definite Clause Grammar	<code>query_parser.pl</code> , <code>items//1</code> (line 58) and <code>goal_phrase//1</code> (line 66).
Higher-order predicates	<code>findall/3</code> , <code>maplist/3</code> , <code>convlist/3</code> , <code>foldl/4</code> for collecting explanations and validating input.

[Line numbers refer to the submitted version of engine.pl and query_parser.pl; re-check them if you edit those files.]

4. Explanation Facility

The brief requires the explanation to show the chain **user input** → **facts** → **rules fired** → **reasoning** → **conclusion**. The system provides this chain at two levels of detail, for users and for examiners, and answers the three classic explanation questions: How, Why and Why not [11], [12].

4.1 How: the reasoning trace

The technical trace is generated from the fired/3 log. For each fired rule it lists the rule's conditions as they were satisfied, the justification and source, and the certainty calculation:

```
USER INPUT
  RI = 1.765 / SG = 4.0 / Optic character = uniaxial / Colour = blue / Inclusion = silk
FACTS (WORKING MEMORY AT START)
  ri(1.765), sg(4.0), optic(uniaxial), observed_colour(blue), inclusion(silk)
RULES FIRED (IN ORDER)
  r1 -> species(corundum)
    because: optic character is uniaxial
      RI 1.765 lies in the corundum range 1.757-1.780 (+/-0.005)
      SG 4.0 lies in the corundum range 3.95-4.10 (+/-0.03)
    why: Corundum is doubly refractive (uniaxial) with RI about 1.762-1.770 and SG about 4.00. [source: gia]
    certainty: rule CF 0.95 x weakest condition 1.00 = 0.95
  r24 -> variety(blue_sapphire)
    because: species(corundum) (established earlier); colour is blue; phenomenon(star) is not established
  r46 -> origin(likely_natural)
    because: species(corundum) (established earlier); inclusion seen: silk
  r61 -> advice(get_certificate)
CONCLUSION
  Blue sapphire (Corundum) - Strong match; Origin: Likely natural
  Recommendation: obtain an NGJA or other accredited laboratory certificate before purchase.
```

The same information is presented to users as numbered plain-language steps (Figure 7), for example "Its refractive index (1.765) is within the Corundum range of 1.757-1.780", each tagged with its rule, certainty and source.

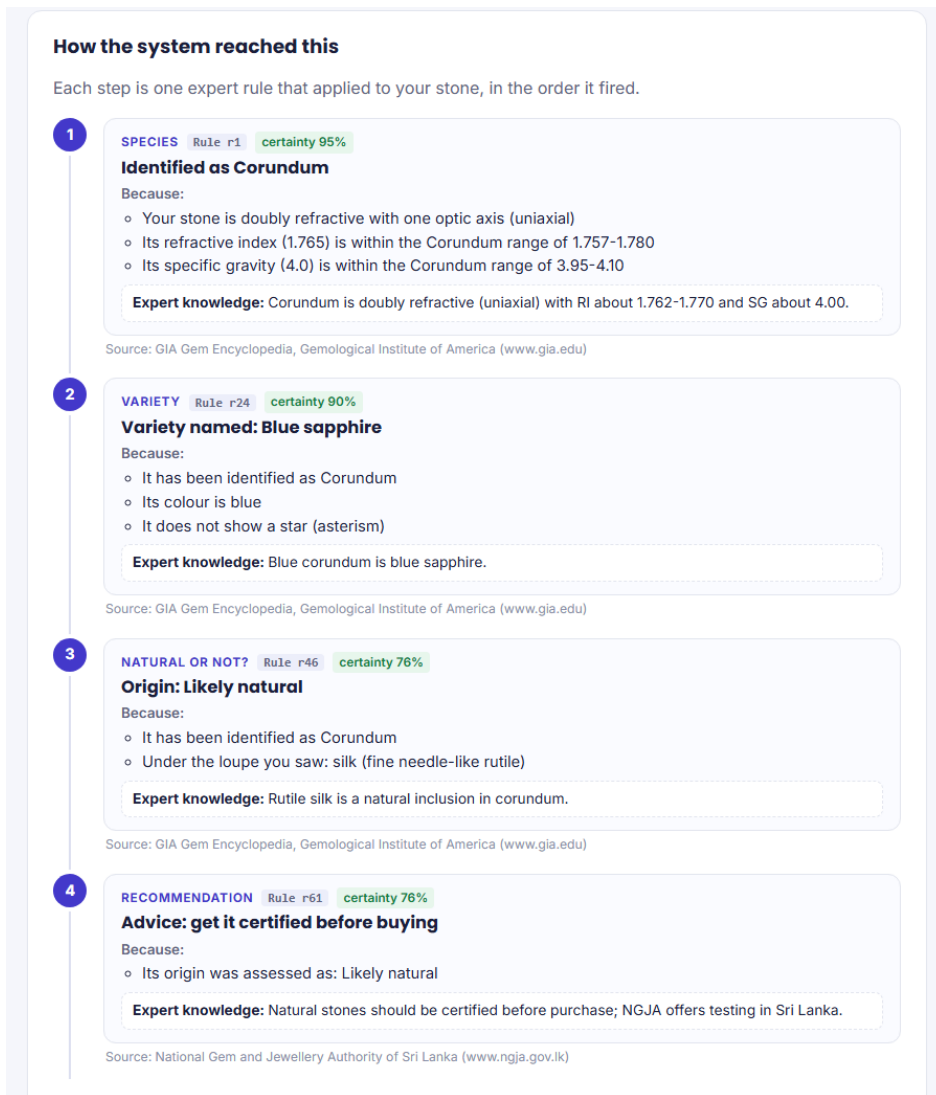


Figure 7. Plain-language How explanation

4.2 Why: explaining a question

During a consultation the user can ask why a question is being asked. The answer is built from the stack of rules being tried (Figure 8):

WHY I AM ASKING:

- I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red AND NOT Effect: Star THEN Variety: Ruby.
- To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
- To test rule r1 I need to know: the optic character.

Assistant [How it works](#) [Identify](#) [Step by step](#) [Examples](#) [Glossary](#) [Knowledge](#)

[← Back to home](#)

Consult mode - interactive backward chaining

Is my stone a Ruby?

The system tests one hypothesis at a time and asks only for the facts the current rule needs.

QUESTION 1 testing rule r1

What is the optic character of your stone?

Optic character
Whether light passes through as one ray or is split in two.

Unknown / not checked

HOW Turn the stone between crossed filters in a polariscope.

Answer **I don't know**

▼ Why are you asking this?

1. I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red AND NOT Effect: Star THEN Variety: Ruby.
2. To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
3. To test rule r1 I need to know: the optic character.

[Start the consultation again](#)

Figure 8. Consultation question with the Why explanation

4.3 Why not: explaining a rejected alternative

When a verification fails, the system names the failed condition and traces it back through the rules. For a red spinel (isotropic, RI 1.718, SG 3.60) checked as a ruby, it answers: *"No - your stone does not match Ruby. Main reason: it must be doubly refractive with one optic axis (uniaxial), but you entered singly refractive (isotropic)"*. It then shows that rule r21 needs corundum and rule r1 failed on the optic character (Figure 9), and adds what the data points to instead (red spinel).

Why not?

The knowledge base has one way to prove a stone is Ruby. Each was checked against your data:

↳ It must first be identified as Corundum, which fails because: Rule r21

✗ It must be doubly refractive with one optic axis (uniaxial), but you entered singly refractive (isotropic) Rule r1

Figure 9. Why-not explanation for a failed verification

4.4 Reasoning diagrams

Each result page includes a diagram generated from the actual reasoning for the user's own input, with Play, Back and Next controls that replay it one step at a time with a caption. The forward-chaining diagram (Figure 10) shows input facts at the top, each fired rule numbered in firing order, and the facts it added. The backward-chaining diagram (Figure 11) shows the goal tree: the question, the rules tried, and each condition marked as proved, failed or never checked, which makes visible that backward chaining abandons a rule at its first failed condition.

Reasoning diagram: forward chaining

Data-driven. The system starts from your inputs (top row) and fires every rule whose conditions are all true. Each fired rule adds a new fact to working memory, which can let further rules fire, until nothing new can be derived. Gold numbers show the firing order. Hover over any box for details.

Play step by step Back Next Show all

Step 5 of 5

Finished: after 4 cycle(s) no rule can add a new fact, so forward chaining stops. The facts at the bottom are the conclusions.

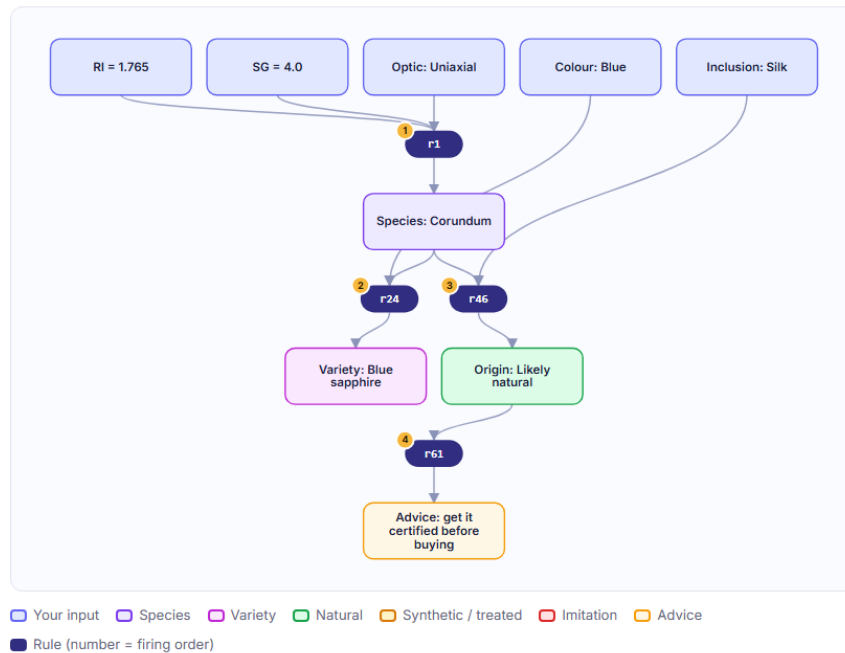


Figure 10. Forward-chaining diagram for a natural blue sapphire

Reasoning diagram: backward chaining

Goal-driven. The system starts from your question (top) and works downwards: it tries each rule that could prove the goal, turning the rule's conditions into sub-goals and checking them from left to right. A rule is abandoned at its first failed condition. Hover over any box for details.

Play step by step Back Next Show all

Step 9 of 9

Result: every rule that could prove it failed, so the answer to "Is it Ruby?" is NO.

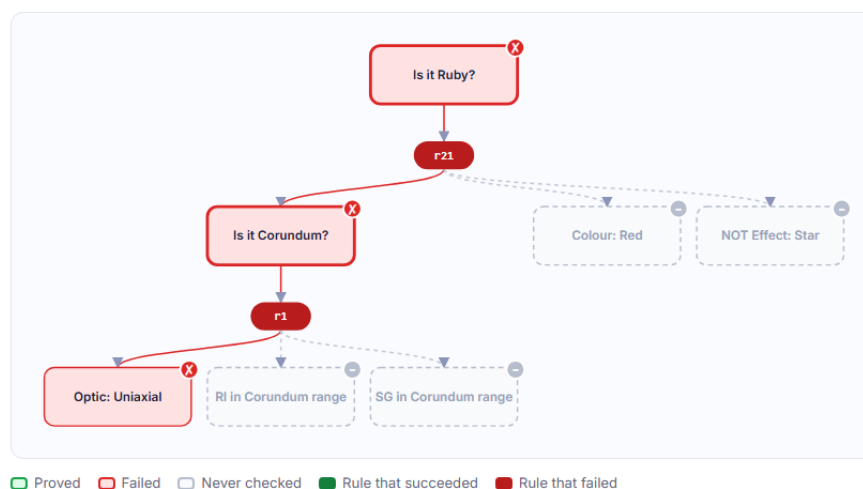


Figure 11. Backward-chaining goal tree for "is it a ruby?" with red spinel data

5. Testing and Evaluation

5.1 Test strategy

Testing was carried out at three levels: (1) documented manual test cases that exercise each type of scenario through the complete system, (2) automated unit tests with SWI-Prolog's plunit library that are re-run after every change, and (3) interface testing of every web page, including invalid input, in a browser. The 14 example stones built into the home page double as a regression set: each was run through both forward and backward chaining and produced its expected result.

5.2 Manual test cases

Table 8 lists the test cases with their expected and actual results. The actual output and fired rules were produced by running the system on each input.

Table 8. Manual test cases

TC	Scenario	Input	Expected output	Actual output	Rules fired	Result
1	Natural blue sapphire	RI 1.765, SG 4.00, uniaxial, blue, silk	Blue sapphire, likely natural, certificate advice	Blue sapphire (90%); origin: Likely natural	r1, r24, r46, r61	Pass
2	Spinel, not garnet	RI 1.718, SG 3.60, isotropic, red	Red spinel; garnet rejected	Red spinel (81%); origin: not determined	r2, r37, r52, r66	Pass
3	Garnet whose SG overlaps corundum	RI 1.79, SG 4.05, isotropic	Garnet, not corundum	Garnet (90%); origin: not determined	r3, r52, r66	Pass
4	Glass imitation	RI 1.60, isotropic, gas bubbles	Imitation: glass	Glass (85%); origin: Imitation	r11, r45, r63	Pass
5	Synthetic ruby	RI 1.765, SG 4.00, uniaxial, red, curved lines	Ruby, synthetic	Ruby (90%); origin: Synthetic	r1, r21, r41, r62	Pass
6	Star sapphire	RI 1.765, SG 4.00, uniaxial, blue, star	Star sapphire (not blue sapphire)	Star sapphire (90%)	r1, r23, r52, r66	Pass
7	Missing RI and SG	isotropic, red	Ranked candidates, measure more	Ranked: Spinel 52%, Garnet 52%	r14, r15, r19, r70	Pass
8	No match	RI 1.40, SG 1.50, uniaxial	Unknown, laboratory test advised	We could not identify this stone	r72	Pass
9	Verify - true	"Is it ruby?" with ruby data	Yes, with proof tree	YES (90%) with proof tree	r21, r1	Pass
10	Verify - false	"Is it spinel?" with sapphire data	No, names failed condition	NO - must be singly refractive (isotropic), but you entered uniaxial	r2 (failed)	Pass
11	Alexandrite	RI 1.748, SG 3.73, biaxial, green, colour change	Alexandrite, confirm-colour-change advice	Alexandrite (86%)	r4, r29, r52, r66, r68	Pass
12	Moonstone	RI 1.52, SG 2.57, biaxial, colourless, adularescence	Moonstone	Moonstone (86%)	r9, r31, r52, r66	Pass
13	Cubic zirconia	SG 5.80, isotropic, colourless	Imitation: cubic zirconia	Cubic zirconia (95%); origin: Imitation	r13, r45, r63	Pass

14	Heat-treated sapphire	RI 1.764, SG 4.00, uniaxial, blue, discoid fractures	Blue sapphire, natural, heated, disclosure advice	Blue sapphire (90%); Likely natural	r1, r24, r49, r50, r61, r67	Pass
15	Ambiguous spinel/garnet	RI 1.733, SG 3.62, isotropic	Both species, re-measure advice	Ranked: Spinel 90%, Garnet 90%	r2, r3, r52, r66, r71	Pass
16	Edge-of-range reading	RI 1.783 (inside tolerance only), SG 4.00, uniaxial, blue	Blue sapphire with reduced certainty	Blue sapphire (63%)	r1, r24, r52, r66	Pass
17	Consultation stops early	Goal ruby; answer optic = isotropic	No, without asking RI, SG or colour	Answer NO after 1 question; RI, SG, colour never asked	r21, r1 (failed)	Pass
18	Plain-English question	"Is my stone a sapphire if the RI is 1.765 and it is blue?"	Goal blue sapphire, facts RI and colour	Parsed as <code>verify(blue_sapphire)</code> with <code>[ri(1.765), observed_colour(blue)]</code>	-	Pass

5.3 Automated tests

The plunit suite in `tests.pl` contains 41 tests in five units (Table 9). All 41 pass (Appendix B). The suite is run with `run_tests`.

Table 9. Automated test suites

Unit	Tests	What is checked
identify_forward_chaining	18	Species, varieties, imitations, origin, treatment, partial data, ambiguity, no match, refractoriness, explanation generation.
verify_backward_chaining	7	Proof trees, failed goals, nested why-not reasons, plain-language explanations.
certainty_factors	5	Rule CF propagation, tolerance penalty, MYCIN combination, ranking, proof certainty.
consultation_why	6	Question order, early stopping, "don't know" answers, hypothesise-and-test, Why text.
natural_language_dcg	5	Verify and identify questions, longest-name matching, multi-word phrases, unknown words.

5.4 Defects found and fixed

Testing found several real defects, which were fixed and, where possible, covered by a regression test (Table 10).

Table 10. Defects found during testing

Defect	Cause	Fix
Evidence rules added almost no certainty ($0.3 \times 0.2 = 0.06$)	An evidence rule's own conclusion (the candidate) was counted as its weakest condition.	Exclude the conclusion from its own weakest-condition calculation.
Consultation page returned "Internal server error"	A nested list of hidden form fields was misread by the HTML generator as attributes.	Flatten the element list; the same fix restored a missing default option in two drop-down menus.
Navigation bar and example gallery collapsed in the browser	Two CSS class names were shared by unrelated elements.	Rename the conflicting classes.

Console menu looped forever when input ended	End of input was treated as menu choice 1.	Treat end of input as "Quit" in menus.
Diagram coordinates were written as unevaluated expressions	Arithmetic was passed to format/2 without evaluation.	Evaluate with is/2 before formatting.

5.5 Evaluation against the objectives

Table 11. Evaluation against the objectives

Objective	Status	Evidence
O1 Knowledge acquisition documented	Met	Table 3; rule_info/3 stores the source of every rule; Appendix A.
O2 At least 20 facts and 20 rules	Exceeded	297 facts, 62 rules in four chained layers.
O3 Forward and backward chaining	Met	Identify, Verify and Consult modes over one rule base; TC1-17.
O4 Explanation facility	Met	How, Why and Why-not at technical and plain-language levels; diagrams.
O5 Incomplete data and confidence	Met	Certainty factors, ranked candidates; TC7, 15, 16.
O6 User-friendly interface	Met	Web and console interfaces, plain-English questions, example stones.
O7 Testing and user manual	Met	18 manual and 41 automated tests; separate User Manual.

The main gap in the evaluation is that all tests use textbook values. The system has not yet been validated against independently certified stones or laboratory reports. Such a validation study, reporting an accuracy table with analysis of any failures, is the most important next step (Section 7.4).

6. User Manual

The installation and operating instructions are provided as a separate document, *Gemstone Identification Assistant - User Manual*. It covers the required software, installation of SWI-Prolog, starting the web and console interfaces, the meaning of each input field and where to obtain it, worked examples with screenshots, the three reasoning modes, troubleshooting, and how to run the automated tests.

7. Discussion

7.1 Modelling expert cognition

The three reasoning modes correspond to recognisable patterns of expert thinking. When a gemmologist first receives an unknown stone, they gather observations and let the evidence suggest conclusions. This is data-driven reasoning, modelled by forward chaining. Once a hypothesis forms ("this could be a sapphire"), expert reasoning becomes **hypothesise-and-test**: the expert selects the tests that would confirm or rule out that hypothesis and abandons it as soon as a test contradicts it. Studies of expert diagnostic reasoning describe the same hypothetico-deductive pattern [18]. The consultation mode models it directly. It forms a hypothesis from the rule order, asks only for the evidence that hypothesis needs, and moves to the next hypothesis on failure. Test case 17 shows the effect: an isotropic reading rules out ruby after a single question.

Human experts also reason with uncertain evidence and weigh several weak clues. The certainty factors model this: no single colour observation identifies a gem, but several weak matches together raise one candidate above another. Finally, experts can justify their conclusions. The explanation facility gives the system the same ability to justify, and its explanations cite the source of each piece of knowledge.

7.2 Symbolic AI compared with machine learning

An alternative would be to train a machine-learning model, for example a neural network that classifies gem photographs [17]. Such models can be very accurate on images and can learn patterns no expert has written down. They are, however, largely "black boxes": they cannot say *why* a stone was classified as a sapphire, they need large labelled data sets that do not exist for Sri Lankan stones and their treatments, and a photograph cannot capture RI, SG or optic character. This rule-based system is transparent. Every conclusion can be traced to named rules and published sources, and it is easy to correct when an expert disagrees. Rudin [19] argues that for high-stakes decisions, interpretable models should be preferred to explaining black boxes after the fact. A buyer deciding whether to spend a large sum on a stone is making exactly that kind of decision, which is why explainability was treated as a core requirement. The weakness of the symbolic approach is the knowledge acquisition bottleneck: every rule must be written and justified by hand, and the system knows nothing outside its rules. A hybrid, in which image recognition proposes candidates that the rule base then checks and explains, is a promising direction.

7.3 Limitations

- **No validation on real certified stones.** Accuracy on real-world readings, with their measurement noise, is unknown.
- **Over-the-limit readings.** Standard refractometers cannot read high zircon or cubic zirconia; the user cannot yet enter "over the limit", so these rely on SG.
- **Unused properties.** Birefringence is stored but not used as an input; pleochroism, fluorescence and absorption spectra, which gemmologists use to separate overlapping species, are not modelled.
- **Treatments.** Only heat treatment is indicated, and only from one visible feature. Glass-filled ruby, a significant fraud, is not covered.
- **One inclusion per stone.** The user can report only the dominant inclusion type.
- **Subjective certainty factors.** The CF values are the knowledge engineer's judgement and have not been calibrated against data. MYCIN-style certainty factors also lack a full probabilistic foundation [17].
- **Rule-order dependence.** Rules with negative conditions rely on the layer order for correct conflict resolution. This is documented but would need care if rules were added carelessly.
- **Consultation diagram.** For "find the species", the diagram shows the successful proof but not the hypotheses rejected before it, although the question sequence and the Why explanations do show them.

7.4 Future work

- Validate on 15-20 certified stones or laboratory reports and report accuracy with an error analysis.
- Add an "over the limit" RI option, birefringence input, and rules for pleochroism, fluorescence and glass-filled ruby.
- Allow several inclusion types per stone, combining their evidence with certainty factors.

- Add a knowledge base consistency checker (conflicting, redundant or unreachable rules) and a knowledge-editing page for experts.
- Explore a hybrid system in which image recognition proposes candidates that the rule base verifies and explains.

8. Conclusion

This project designed and implemented a rule-based expert system that identifies common Sri Lankan gemstones, detects synthetics and imitations, and explains its reasoning. Knowledge from recognised gemmological sources was systematically acquired, documented and represented as 297 facts and 62 production rules in four chained layers, with a certainty factor and a literature source for every rule. One rule base supports three reasoning modes: forward chaining for identification, backward chaining for verifying a claim, and an interactive consultation that asks only for the facts it needs. A full How, Why and Why-not explanation facility, reasoning diagrams, a plain-English question interface and a web interface make the reasoning transparent to non-experts. All 18 documented test cases and 41 automated tests pass. The system meets its objectives as a tool for a first opinion. Its main limitations, the absence of real-world validation and several unmodelled gemmological properties, define clear next steps. Above all, the project shows why a transparent, explainable system is appropriate when a person's money depends on its answer.

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[Check each reference, especially edition numbers and years, against the copy you actually used.]

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Appendix A. Complete Rule Base

Generated from knowledge_base.pl. Source keys: gia [1], webster [2], read [3], matlins [4], igs [5], gemdat [6], ngja [7], design = knowledge engineer decision.

Id	IF (conditions)	THEN	CF	Justification [source]
r1	optic(uniaxial) AND ri_in(corundum) AND sg_in(corundum)	species(corundum)	0.95	Corundum is doubly refractive (uniaxial) with RI about 1.762-1.770 and SG about 4.00. [gia]
r2	optic(isotropic) AND ri_in(spinel) AND sg_in(spinel)	species(spinel)	0.90	Spinel is singly refractive (isotropic) with RI about 1.718 and SG about 3.60. [webster]
r3	optic(isotropic) AND ri_in(garnet) AND sg_in(garnet)	species(garnet)	0.90	Garnets are singly refractive with RI 1.73-1.89 and SG 3.6-4.3 depending on the type. [webster]
r4	optic(biaxial) AND ri_in(chrysoberyl) AND sg_in(chrysoberyl)	species(chrysoberyl)	0.95	Chrysoberyl is biaxial with RI about 1.746-1.755 and SG about 3.73. [gia]
r5	optic(uniaxial) AND ri_in(zircon) AND sg_in(zircon)	species(zircon)	0.90	High zircon is uniaxial with very high RI (often over the refractometer limit) and SG about 4.7. [gemdat]
r6	optic(biaxial) AND ri_in(topaz) AND sg_in(topaz)	species(topaz)	0.95	Topaz is biaxial with RI about 1.61-1.64 and SG about 3.53. [gia]
r7	optic(uniaxial) AND ri_in(tourmaline) AND sg_in(tourmaline)	species(tourmaline)	0.90	Tourmaline is uniaxial with RI about 1.62-1.64 and SG about 3.06. [gia]
r8	optic(uniaxial) AND ri_in(quartz) AND sg_in(quartz)	species(quartz)	0.90	Quartz is uniaxial with RI 1.544-1.553 and SG 2.65. [read]
r9	optic(biaxial) AND ri_in(feldspar) AND sg_in(feldspar)	species(feldspar)	0.90	Orthoclase feldspar (moonstone) is biaxial with RI about 1.52 and SG about 2.57. [read]
r10	optic(uniaxial) AND ri_in(beryl) AND sg_in(beryl)	species(beryl)	0.90	Beryl is uniaxial with RI about 1.57-1.60 and SG about 2.7. [gemdat]
r11	optic(isotropic) AND ri_in(glass) AND inclusion(gas_bubbles) AND not(species(G))	imitation(glass)	0.85	A singly refractive stone with a glass RI and round gas bubbles, matching no gem species, is glass. [igs]
r12	inclusion(swirl_marks) AND not(species(G))	imitation(glass)	0.80	Swirl marks (flow lines) are typical of moulded glass. [matlins]
r13	optic(isotropic) AND sg_in(cubic_zirconia)	imitation(cubic_zirconia)	0.95	Cubic zirconia is isotropic with an SG of 5.6-6.0, far heavier than any natural gem in scope. [gia]

r14	partial_data AND not(species(G)) AND not(imitation(H)) AND gem_type(I) AND optic_ok(I) AND ri_ok(I) AND sg_ok(I) AND colour_ok(I) AND phenomenon_ok(I)	candidate(I)	0.20	When RI, SG or optic character is missing, every gem consistent with ALL the supplied properties remains a candidate. [design]
r15	candidate(G) AND not(missing(optic)) AND optic_ok(G)	candidate(G)	0.30	A matching optic character is moderate evidence for a candidate. [design]
r16	candidate(G) AND not(missing(ri)) AND ri_ok(G)	candidate(G)	0.50	A matching refractive index is the strongest single test in gemmology. [read]
r17	candidate(G) AND not(missing(sg)) AND sg_ok(G)	candidate(G)	0.40	A matching specific gravity is good supporting evidence. [read]
r18	candidate(G) AND not(missing(phenomenon)) AND not(phenomenon(none)) AND phenomenon_ok(G)	candidate(G)	0.40	An optical effect the species is known for is good evidence. [gia]
r19	candidate(G) AND typical_colour_ok(G)	candidate(G)	0.15	A colour that is typical for the species is weak supporting evidence. [matlins]
r21	species(corundum) AND observed_colour(red) AND not(phenomenon(star))	variety(ruby)	0.95	Red corundum is called ruby. [gia]
r22	species(corundum) AND observed_colour(red) AND phenomenon(star)	variety(star_ruby)	0.95	Red corundum showing asterism (a star) is star ruby. [gia]
r23	species(corundum) AND phenomenon(star) AND not(observed_colour(red))	variety(star_sapphire)	0.95	Non-red corundum showing asterism is star sapphire. [gia]
r24	species(corundum) AND observed_colour(blue) AND not(phenomenon(star))	variety(blue_sapphire)	0.95	Blue corundum is blue sapphire. [gia]
r25	species(corundum) AND observed_colour(pink) AND not(phenomenon(star))	variety(pink_sapphire)	0.85	Pink corundum (lighter than ruby) is sold as pink sapphire. [gia]
r26	species(corundum) AND observed_colour(orange) AND not(phenomenon(star))	variety(padparadscha)	0.70	Pinkish-orange to orange corundum from Sri Lanka is traded as padparadscha. [gia]
r27	species(corundum) AND observed_colour(yellow) AND not(phenomenon(star))	variety(yellow_sapphire)	0.90	Yellow corundum is yellow sapphire. [gia]
r28	species(corundum) AND observed_colour(colourless) AND not(phenomenon(star))	variety(white_sapphire)	0.90	Colourless corundum is white sapphire. [gia]
r29	species(chrysoberyl) AND phenomenon(colour_change)	variety(alexandrite)	0.90	Chrysoberyl that changes colour between daylight and incandescent light is alexandrite. [gia]
r30	species(chrysoberyl) AND phenomenon(cats_eye)	variety(cats_eye_chrysoberyl)	0.95	Chrysoberyl showing chatoyancy is cat's-eye

				chrysoberyl (cymophane). [gia]
r31	species(feldspar) AND phenomenon(adularescence)	variety(moonstone)	0.95	Orthoclase feldspar showing adularescence is moonstone. [gia]
r32	species(quartz) AND observed_colour(purple)	variety(amethyst)	0.95	Purple quartz is amethyst. [gia]
r33	species(quartz) AND observed_colour(yellow)	variety(citrine)	0.90	Yellow quartz is citrine. [gia]
r34	species(garnet) AND observed_colour(orange)	variety(hessonite)	0.85	Orange to brownish-orange garnet is hessonite (grossular). [gia]
r35	species(garnet) AND observed_colour(red)	variety(pyrope_almandine)	0.80	Red garnet is usually of the pyrope-almandine series. [webster]
r36	species(beryl) AND observed_colour(blue)	variety(aquamarine)	0.90	Blue beryl is aquamarine. [gia]
r37	species(spinel) AND observed_colour(red)	variety(red_spinel)	0.90	Red spinel is a valued variety often confused with ruby. [gia]
r38	species(zircon) AND observed_colour(blue)	variety(blue_zircon)	0.85	Blue zircon is almost always produced by heating brown zircon. [gia]
r41	species(corundum) AND inclusion(curved_lines)	origin(synthetic)	0.95	Curved growth lines (striae) are diagnostic of flame-fusion (Verneuil) synthetic corundum; natural corundum grows in straight lines. [gia]
r42	species(corundum) AND inclusion(gas_bubbles)	origin(synthetic)	0.90	Round gas bubbles in corundum indicate a flame-fusion synthetic. [gia]
r43	species(spinel) AND inclusion(gas_bubbles)	origin(synthetic)	0.85	Gas bubbles in spinel indicate flame-fusion synthetic spinel. [webster]
r44	inclusion(gas_bubbles) AND not(species(G)) AND not(imitation(H))	origin(glass_or_synthetic)	0.70	Round gas bubbles in an unidentified stone indicate glass or a synthetic. [igs]
r45	imitation(G)	origin(imitation)	1.00	An identified imitation material is not a natural gemstone. [design]
r46	species(corundum) AND inclusion(silk)	origin(likely_natural)	0.80	Rutile silk is a natural inclusion in corundum. [gia]
r47	species(G) AND inclusion(crystals)	origin(likely_natural)	0.70	Included mineral crystals are a typical sign of natural growth. [matlins]
r48	species(G) AND inclusion(fingerprints)	origin(likely_natural)	0.60	Fingerprints (healed fractures) are usually natural, although flux synthetics can show similar features. [matlins]

r49	species(corundum) AND inclusion(discoïd_fractures)	treatment(likely_heated)	0.80	Discoïd fractures around crystals form when corundum is heated. [gia]
r50	treatment(likely_heated)	origin(likely_natural)	0.90	Heat treatment is applied to natural stones, so a heated stone is natural (but treated). [gia]
r51	species(corundum) AND inclusion(none)	origin(needs_lab_test)	0.60	Natural corundum is rarely inclusion-free; a clean stone needs a lab test to rule out a synthetic. [igs]
r52	species(G) AND missing(inclusion)	origin(undetermined)	1.00	Without inclusion evidence the origin cannot be decided. [design]
r53	species(G) AND inclusion(none) AND not(species(corundum))	origin(undetermined)	1.00	A clean stone of this species gives no origin evidence at loupe level. [design]
r61	origin(likely_natural)	advice(get_certificate)	1.00	Natural stones should be certified before purchase; NGJA offers testing in Sri Lanka. [ngja]
r62	origin(synthetic)	advice(synthetic_price)	1.00	Synthetic gems are far cheaper than natural ones and must be sold as synthetic. [igs]
r63	origin(imitation)	advice(imitation_value)	1.00	Imitations have little gem value. [igs]
r64	origin(glass_or_synthetic)	advice(lab_test_bubbles)	1.00	Stones showing gas bubbles need laboratory confirmation. [igs]
r65	origin(needs_lab_test)	advice(lab_test_clean)	1.00	Clean corundum requires advanced laboratory tests to prove natural origin. [gia]
r66	origin(undetermined)	advice(check_inclusions)	1.00	Inclusions seen with a 10x loupe are the main field test for origin. [matlins]
r67	treatment(likely_heated)	advice(declare_heat)	1.00	Treatments must be disclosed to the buyer. [ngja]
r68	variety(alexandrite)	advice(check_colour_change)	1.00	Alexandrite is identified by its daylight/incandescent colour change. [gia]
r69	variety(padparadscha)	advice(padparadscha_cert)	1.00	The padparadscha name is reserved for a specific colour range. [gia]
r70	candidate(G)	advice(measure_more)	1.00	Additional measurements are needed to narrow down the candidates. [design]
r71	ambiguous_species	advice(re_measure)	1.00	Overlapping readings must be resolved by more precise measurement. [design]

r72	not(species(G)) AND not(candidate(H)) AND not(imitation(I))	advice(no_match)	1.00	When no rule identifies the stone, the system must not guess. [design]
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Appendix B. Automated Test Output

```
% Start unit: identify_forward_chaining
% [1/41] identify_forward..tural_blue_sapphire ..... passed (0.016 sec)
% [2/41] identify_forward..2_spinel_not_garnet ..... passed (0.000 sec)
% [3/41] identify_forward..garnet_not_corundum ..... passed (0.000 sec)
% [4/41] identify_forward..c04_glass_imitation ..... passed (0.000 sec)
% [5/41] identify_forward..tc05_synthetic_ruby ..... passed (0.000 sec)
% [6/41] identify_forward...tc06_star_sapphire ..... passed (0.000 sec)
% [7/41] identify_forward..ri_gives_candidates ..... passed (0.000 sec)
% [8/41] identify_forward..ining:tc08_no_match ..... passed (0.000 sec)
% [9/41] identify_forward..ng:tc11_alexandrite ..... passed (0.000 sec)
% [10/41] identify_forward..ning:tc12_moonstone .... passed (0.000 sec)
% [11/41] identify_forward..tc13_cubic_zirconia .... passed (0.000 sec)
% [12/41] identify_forward..c14_heated_sapphire .... passed (0.000 sec)
% [13/41] identify_forward..active_unknown_axis .... passed (0.000 sec)
% [14/41] identify_forward..guous_spinel_garnet .... passed (0.000 sec)
% [15/41] identify_forward...corundum_needs_lab .... passed (0.000 sec)
% [16/41] identify_forward..etic_spinel_bubbles .... passed (0.000 sec)
% [17/41] identify_forward..for_same_conclusion .... passed (0.000 sec)
% [18/41] identify_forward..nation_is_generated .... passed (0.003 sec)
% End unit identify_forward_chaining: passed (0.016 sec CPU)
% Start unit: verify_backward_chaining
% [19/41] verify_backward_c..09_verify_ruby_true .... passed (0.000 sec)
% [20/41] verify_backward_c..es_failed_condition .... passed (0.000 sec)
% [21/41] verify_backward_c..false_nested_reason .... passed (0.000 sec)
% [22/41] verify_backward_c..g:tc22_verify_glass .... passed (0.000 sec)
% [23/41] verify_backward_c..fy_explanation_text .... passed (0.000 sec)
% [24/41] verify_backward_c..teps_plain_language .... passed (0.001 sec)
% [25/41] verify_backward_c..why_not_main_reason .... passed (0.000 sec)
% End unit verify_backward_chaining: passed (0.000 sec CPU)
% Start unit: certainty_factors
% [26/41] certainty_factors..pecies_cf_from_rule .... passed (0.000 sec)
% [27/41] certainty_factors..e_reading_lowers_cf .... passed (0.000 sec)
% [28/41] certainty_factors..ombines_mycin_style .... passed (0.001 sec)
% [29/41] certainty_factors...most_certain_first .... passed (0.002 sec)
% [30/41] certainty_factors:tc30_proof_certainty .... passed (0.000 sec)
% End unit certainty_factors: passed (0.000 sec CPU)
% Start unit: consultation_why
% [31/41] consultation_why...ptic_first_for_ruby .... passed (0.000 sec)
% [32/41] consultation_why...en_hypothesis_fails .... passed (0.000 sec)
% [33/41] consultation_why...ks_next_needed_fact .... passed (0.000 sec)
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```
% [34/41] consultation_why:..se_and_test_species .... passed (0.000 sec)
% [35/41] consultation_why:..t_know_is_respected .... passed (0.000 sec)
% [36/41] consultation_why:..anation_names_rules .... passed (0.000 sec)
% End unit consultation_why: passed (0.000 sec CPU)
% Start unit: natural_language_dcg
% [37/41] natural_language_..c37_verify_question .... passed (0.000 sec)
% [38/41] natural_language_..dentify_description .... passed (0.000 sec)
% [39/41] natural_language_..ngest_gem_name_wins .... passed (0.000 sec)
% [40/41] natural_language_..0_multiword_phrases .... passed (0.000 sec)
% [41/41] natural_language_..nown_words_reported .... passed (0.000 sec)
% End unit natural_language_dcg: passed (0.000 sec CPU)
% All 41 tests passed in 0.064 seconds (0.031 cpu)
```

Appendix C. Console Consultation Transcript

Consult mode, goal "find the species". The user types *why* at the first question, then answers isotropic, RI 1.718 and SG 3.60.

What is the optic character?

- 0. Don't know
 - 1. Isotropic
 - 2. Uniaxial
 - 3. Biaxial
 - 4. Doubly refractive
- (type a number, or WHY)

> WHY I AM ASKING:

- I am trying to find out: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
- To test rule r1 I need to know: the optic character.

What is the optic character?

- 0. Don't know
 - 1. Isotropic
 - 2. Uniaxial
 - 3. Biaxial
 - 4. Doubly refractive
- (type a number, or WHY)

>

What is the refractive index (RI)?

(type a number, press Enter if you don't know, or type WHY)

>

What is the specific gravity (SG)?

(type a number, press Enter if you don't know, or type WHY)

>

Backward chaining asked 3 of the 6 possible questions.

YES - your stone is consistent with Spinel (certainty 90%).
Every condition the expert rules require is met.

HOW IT WAS CHECKED

1. [Species] Identified as Spinel (rule r2; certainty 90%)

- Your stone is singly refractive (isotropic)
- Its refractive index (1.718) is within the Spinel range of 1.712-1.736
- Its specific gravity (3.6) is within the Spinel range of 3.55-3.70

Expert knowledge: Spinel is singly refractive (isotropic) with RI about 1.718 and SG about 3.60.